Abstract

This report presents a design for a 1-bit NAND-based full-adder using 180nm technology. The NAND gate and the full adder schematics and layouts were simulated and their propagation delay and power consumption were compared with those reported in the reference design in [1]. The power consumption closely matched the reference design, but parasitics from the layout design produced significantly larger propagation delays. The causes of this performance loss are identified and discussed, and layout adjustments are suggested to improve performance.

1 - Introduction

A 1-bit full-adder is a circuit that adds two 1-bit inputs and a carry-in to produce a one bit sum and a carry bit. Full adders can be cascaded to perform addition on multi-bit inputs. Addition is a fundamental arithmetic operation; in ALUs, adder circuitry is used to perform addition and subtraction, and supports stages of multiplication and division operations.

Table 1: Truth Table for 1-bit Full Adder

A	B	Cin	Cout	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0

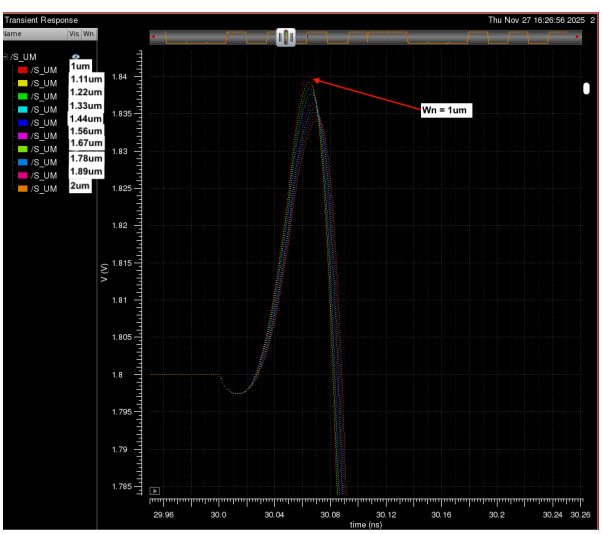
1	1	0	1	0
1	1	1	1	1

The design objectives for this project were to achieve similar propagation delay and power consumption to the NAND gate and NAND-based Full Adder in the 180nm Ripple Carry Adder designed in [1].

2 - Description of the Design

2.1 - NAND Gate Design

The design process began with the CMOS NAND gate schematic. To achieve similar performance to the reference design in [1], the four transistors were sized according to their recommendations: minimum length, with $W_n = 1\mu m$ and $W_p = 1.7\mu m$. The two NMOS devices in the pull-down network were identically sized, as were the two PMOS devices in the pull-up network. The full adder was simulated with different MOSFET gate widths to understand the sizing presented in the NAND gate in [1] and verify the impact of gate width on delay and power dissipation.



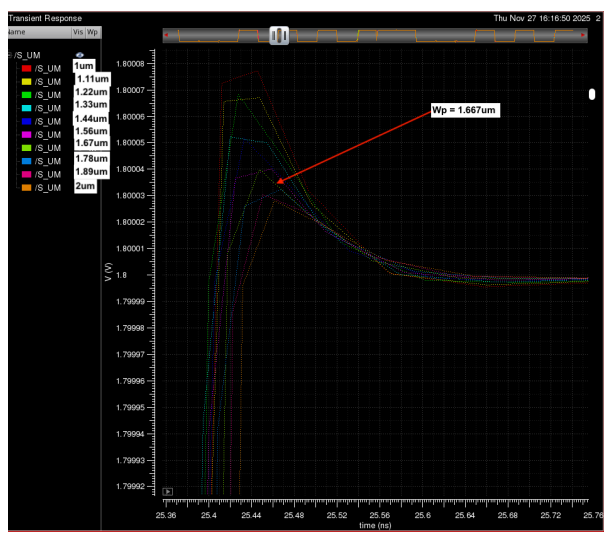


Figure 1: Effect of NMOS Gate Width on Adder Sum Output Fall Time

Figure 2: Effect of PMOS Gate Width on Adder Sum Output Rise Time

Increasing the pull-up and pull-down transistor widths in a NAND gate reduces its propagation delay, but also increases each transistor's gate capacitance ($C_g = W \times L \times C_{ox}$) and consequently the gate's total input capacitance. This tradeoff becomes significant in the full adder design, where two of the NAND gates drive three different inputs; this high fan-out increases their capacitive load, making gate width a significant factor in propagation delay and dynamic power dissipation at these stages.

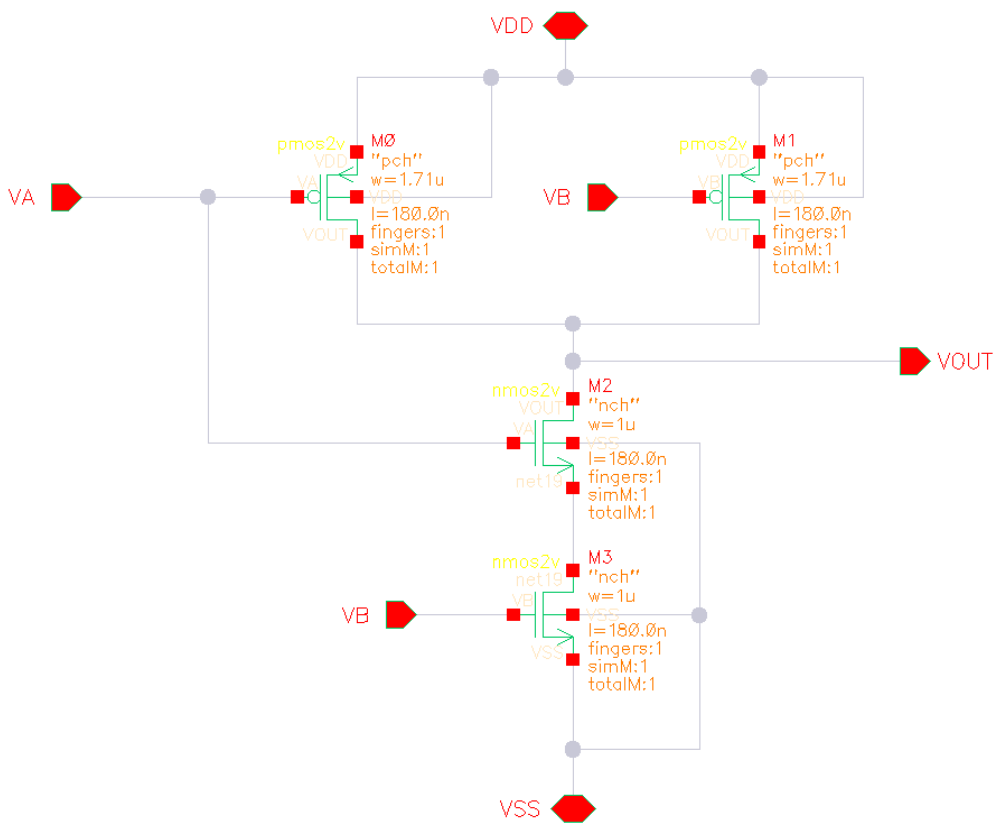


Figure 3: CMOS NAND Gate Schematic

The NAND gate was implemented using CMOS logic. The CMOS configuration ensures consistent voltage levels and low static power dissipation. In this configuration, the 2 parallel PMOS in the pull-up network charge the output node when either input is ‘0’. The 2 series NMOS in the pull-down network discharge the output node when both inputs are ‘1’.

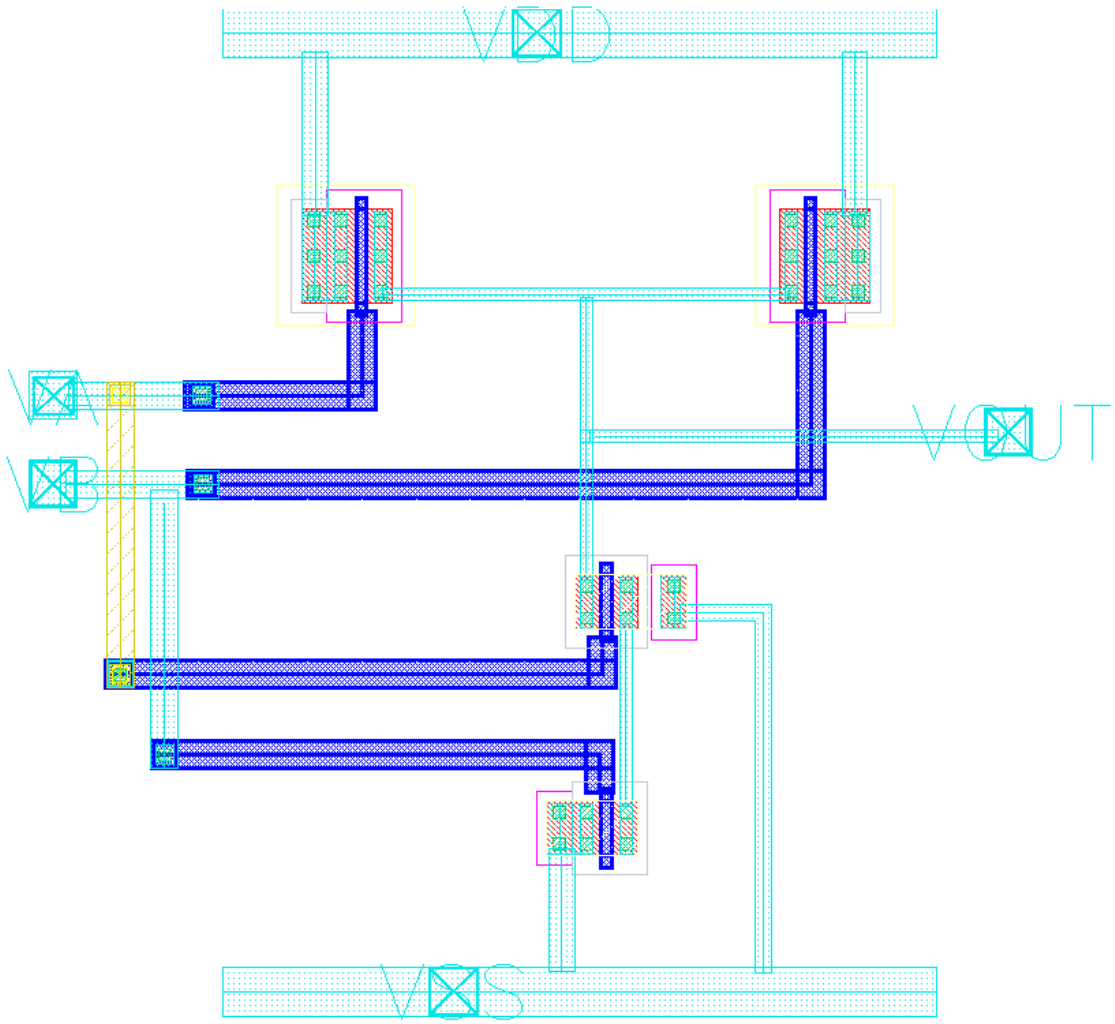
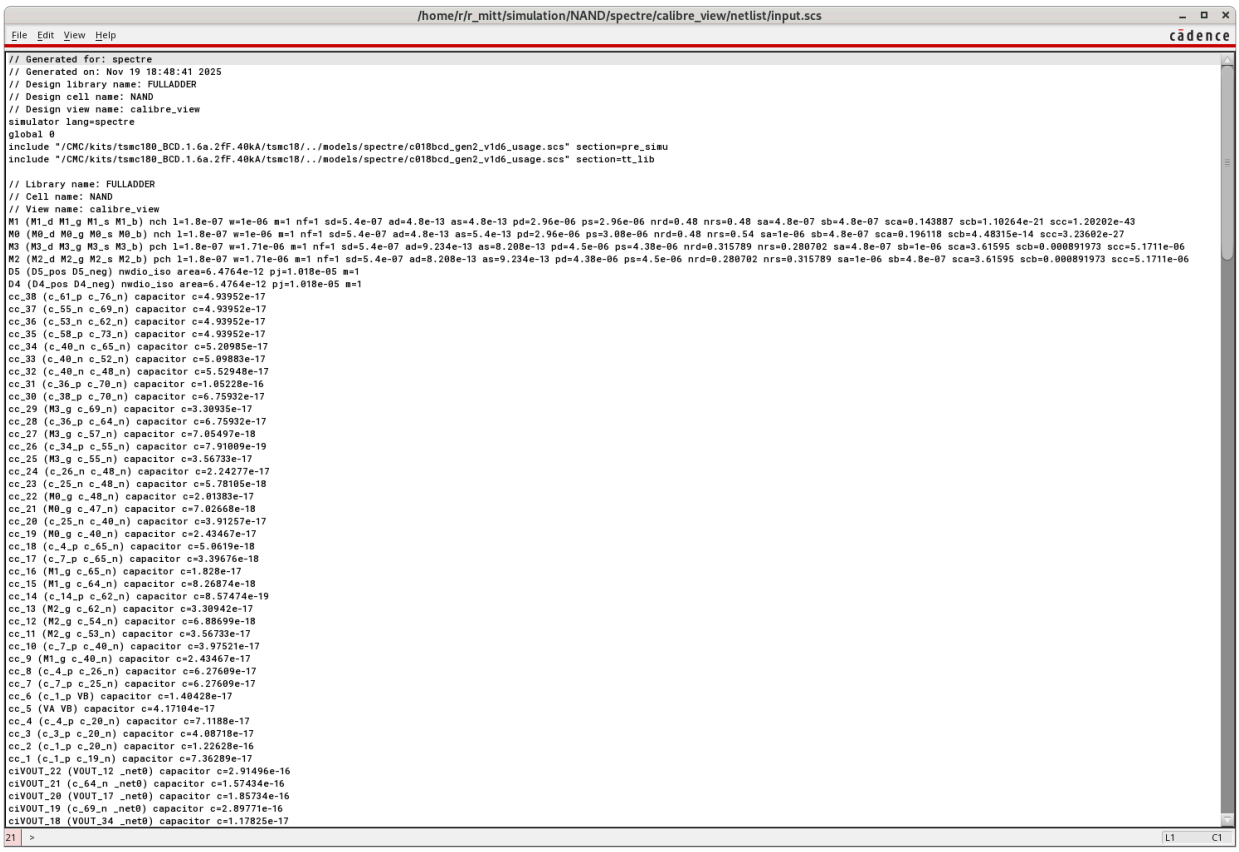


Figure 4: NAND Gate Layout

The extracted layout of the NAND gate introduces several parasitic effects that influence its performance relative to the schematic-level design. The Calibre PEX results (see Figure 5) indicate that the largest parasitic capacitances occur at the output node, where diffusion from the two parallel PMOS devices and the upper NMOS device meet. Additional metal-to-metal and metal-to-poly capacitances arise from the horizontal routing required to bring the VOUT line to the cell boundary. These parasitic elements increase the effective load seen by the driving stages and therefore contribute to longer rise and fall times compared to the ideal NAND schematic.



/home/r/r_mitt/simulation/NAND/spectre/calibre_view/netlist/input.scs

File Edit View Help

```
// Generated for: spectre
// Generated on: Nov 19 18:48:41 2025
// Design library name: FULLADDER
// Design cell name: NAND
// Design view name: calibre_view
simulator lang=spectre
global 0
include "/CMC/kits/tsmc180_BCD.1.6a.2fF.40kA/tsmc18/../models/spectre/c018bcd_gen2_v1d6_usage.scs" section=pre_simu
include "/CMC/kits/tsmc180_BCD.1.6a.2fF.40kA/tsmc18/../models/spectre/c018bcd_gen2_v1d6_usage.scs" section=tt_lib

// Library name: FULLADDER
// Cell name: NAND
// View name: calibre_view
M1 (M1_d M1_g M1_s M1_b) nch l=1.8e-07 w=1e-06 m=1 nf=1 sd=5.4e-07 ad=4.8e-13 as=4.8e-13 pd=2.96e-06 ps=2.96e-06 nrd=0.48 nrs=0.48 sa=4.8e-07 sb=4.8e-07 sca=0.143887 scb=1.10264e-21 scc=1.20202e-43
M0 (M0_d M0_g M0_s M0_b) nch l=1.8e-07 w=1e-06 m=1 nf=1 sd=5.4e-07 ad=4.8e-13 as=5.4e-13 pd=2.96e-06 ps=3.08e-06 nrd=0.48 nrs=0.54 sa=1e-06 sb=4.8e-07 sca=0.196118 scb=4.48315e-14 scc=3.23602e-27
M3 (M3_d M3_g M3_s M3_b) pch l=1.8e-07 w=1.71e-06 m=1 nf=1 sd=5.4e-07 ad=9.234e-13 as=8.208e-13 pd=4.5e-06 ps=4.38e-06 nrd=0.315789 nrs=0.280702 sa=4.8e-07 sb=1e-06 sca=3.61595 scb=0.000891973 scc=5.1711e-06
M2 (M2_d M2_g M2_s M2_b) pch l=1.8e-07 w=1.71e-06 m=1 nf=1 sd=5.4e-07 ad=8.208e-13 as=9.234e-13 pd=4.38e-06 ps=4.5e-06 nrd=0.280702 nrs=0.315789 sa=1e-06 sb=4.8e-07 sca=3.61595 scb=0.000891973 scc=5.1711e-06
D5 (D5_pos D5_neg) nwdio_iso area=6.4764e-12 pj=1.018e-05 m=1
D4 (D4_pos D4_neg) nwdio_iso area=6.4764e-12 pj=1.018e-05 m=1
cc_38 (c_61_p c_76_n) capacitor c=4.93952e-17
cc_37 (c_55_n c_69_n) capacitor c=4.93952e-17
cc_36 (c_53_n c_62_n) capacitor c=4.93952e-17
cc_35 (c_58_p c_73_n) capacitor c=4.93952e-17
cc_34 (c_40_n c_65_n) capacitor c=5.20985e-17
cc_33 (c_40_n c_52_n) capacitor c=5.09883e-17
cc_32 (c_40_n c_48_n) capacitor c=5.52948e-17
cc_31 (c_36_p c_70_n) capacitor c=1.05228e-16
cc_30 (c_38_p c_70_n) capacitor c=6.75932e-17
cc_29 (M3_g c_69_n) capacitor c=3.30935e-17
cc_28 (c_36_p c_64_n) capacitor c=6.75932e-17
cc_27 (M3_g c_57_n) capacitor c=7.05497e-18
cc_26 (c_34_p c_55_n) capacitor c=7.91009e-19
cc_25 (M3_g c_55_n) capacitor c=3.56733e-17
cc_24 (c_26_n c_48_n) capacitor c=2.24277e-17
cc_23 (c_25_n c_48_n) capacitor c=5.78105e-18
cc_22 (M0_g c_48_n) capacitor c=2.01383e-17
cc_21 (M0_g c_47_n) capacitor c=7.02668e-18
cc_20 (c_25_n c_40_n) capacitor c=3.91257e-17
cc_19 (M0_g c_40_n) capacitor c=2.43467e-17
cc_18 (c_4_p c_65_n) capacitor c=5.0619e-18
cc_17 (c_7_p c_65_n) capacitor c=3.39676e-18
cc_16 (M1_g c_65_n) capacitor c=1.828e-17
cc_15 (M1_g c_64_n) capacitor c=8.26874e-18
cc_14 (c_14_p c_62_n) capacitor c=8.57474e-19
cc_13 (M2_g c_62_n) capacitor c=3.30942e-17
cc_12 (M2_g c_54_n) capacitor c=6.88699e-18
cc_11 (M2_g c_53_n) capacitor c=3.56733e-17
cc_10 (c_7_p c_40_n) capacitor c=3.97521e-17
cc_9 (M1_g c_40_n) capacitor c=2.43467e-17
cc_8 (c_4_p c_26_n) capacitor c=6.27609e-17
cc_7 (c_7_p c_25_n) capacitor c=6.27609e-17
cc_6 (c_1_p VB) capacitor c=1.40428e-17
cc_5 (VA VB) capacitor c=4.17104e-17
cc_4 (c_4_p c_20_n) capacitor c=7.1188e-17
cc_3 (c_3_p c_20_n) capacitor c=4.08718e-17
cc_2 (c_1_p c_20_n) capacitor c=1.22628e-16
cc_1 (c_1_p c_19_n) capacitor c=7.36289e-17
ciVOUT_22 (VOUT_12 _net0) capacitor c=2.91496e-16
ciVOUT_21 (c_64_n _net0) capacitor c=1.57434e-16
ciVOUT_20 (VOUT_17 _net0) capacitor c=1.85734e-16
ciVOUT_19 (c_69_n _net0) capacitor c=2.89771e-16
ciVOUT_18 (VOUT_34 _net0) capacitor c=1.17825e-17
```

Figure 5: NAND PEX Extracted Netlist

The input paths for A and B are routed using short Metal 1 segments for local gate connections, while the V_A input also includes a short Metal 2 run to reach the NMOS gate. This higher-level routing avoids congestion in the active device region and keeps the interconnect length minimal, resulting in limited additional parasitic RC. Internally,

the node between the two series NMOS devices is kept compact, with short local interconnects to reduce capacitance at the switching node. Since this node participates in conduction only when both inputs are high, minimizing its parasitic loading helps reduce the falling-edge delay of the NAND gate.

In this layout, diffusion regions are not shared between transistors in the pull-up or pull-down networks. Although the schematic topology of the NAND gate would allow shared diffusion between the series NMOS devices or between the parallel PMOS devices, each transistor is instead implemented using its own isolated diffusion region, connected to VDD, VSS, or the internal node through metal. This simplifies the layout and routing but results in slightly larger junction capacitances and a marginally larger active area compared to a diffusion-shared implementation.

The final layout is area-efficient and follows a standard cell-style structure. The PMOS network is placed near the VDD rail and the NMOS network near the VSS rail, with both rails implemented as wide horizontal metal tracks spanning the width of the cell. This fixed top and bottom rail arrangement, along with the confinement of all devices within the VDD–VSS vertical height, mirrors the structure of CMOS standard cells and allows the NAND to be placed adjacent to other gates without violating routing or power distribution constraints. The remainder of the layout uses only the required via stacks for pin access and for transitions between poly, Metal 1, and Metal 2 (e.g., the VA input connection). By aligning the rails with the upper and lower cell boundaries, the NAND gate can be tiled efficiently with other cells in the full-adder layout while maintaining continuity in the power distribution network.

After completing the layout, design rule checking (DRC) was performed to validate manufacturability. The NAND gate passed all mandatory rules, confirming proper spacing, enclosure, and layer interactions. Likewise, layout-versus-schematic (LVS) verification confirmed that the physical layout matches the intended circuit connectivity with no mismatches reported. (Results shown in Appendix A.)

2.2 - Full Adder Design

The full-adder uses 9 NAND gates to generate the sum and carry outputs.

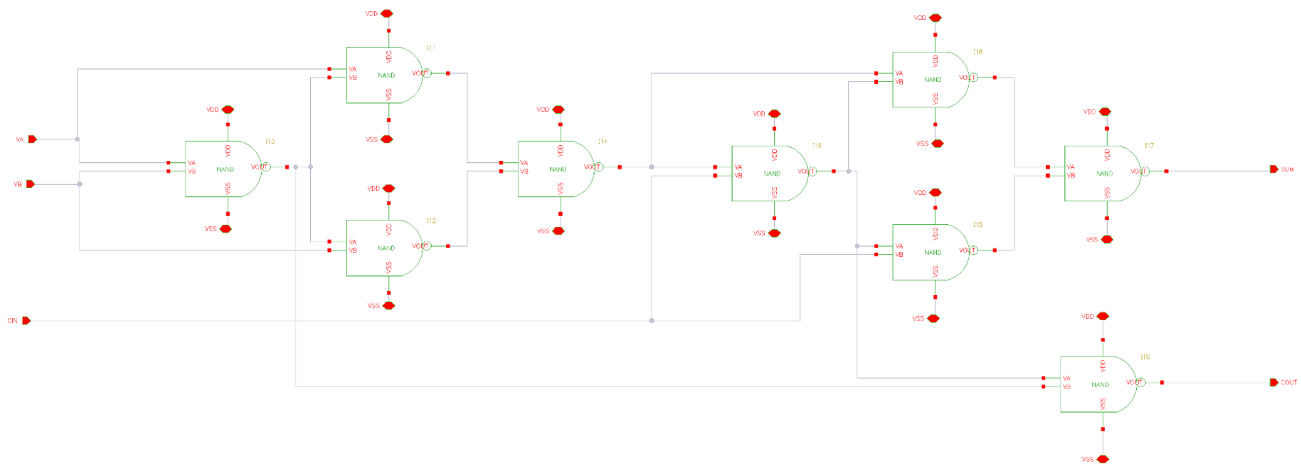


Figure 6: NAND-based Full-Adder Schematic

The sum path has the greatest number of cascaded NAND stages, requiring the signal to propagate through several gates before reaching the Sum node. The Sum output is therefore expected to have the longest propagation delay in this design.

The outputs of gates 1 and 5 (see Figure 7) have the highest fan-out; each drives 3 gate inputs. This higher capacitive load will increase propagation delay and will result in higher dynamic power dissipation at these nodes.

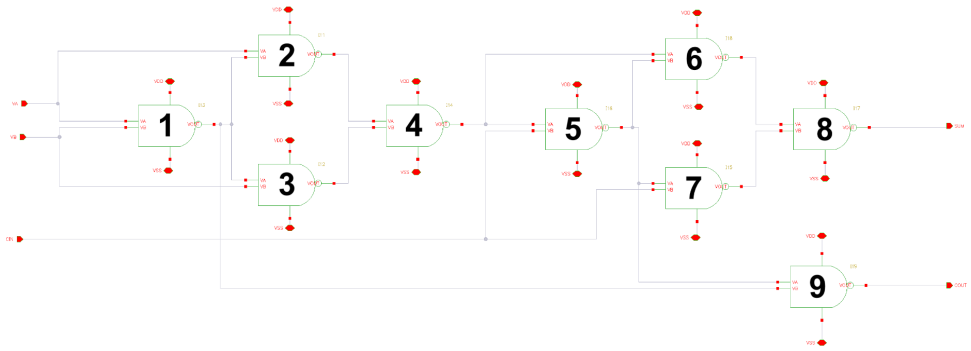


Figure 7: Full Adder Schematic with Numbered Gates

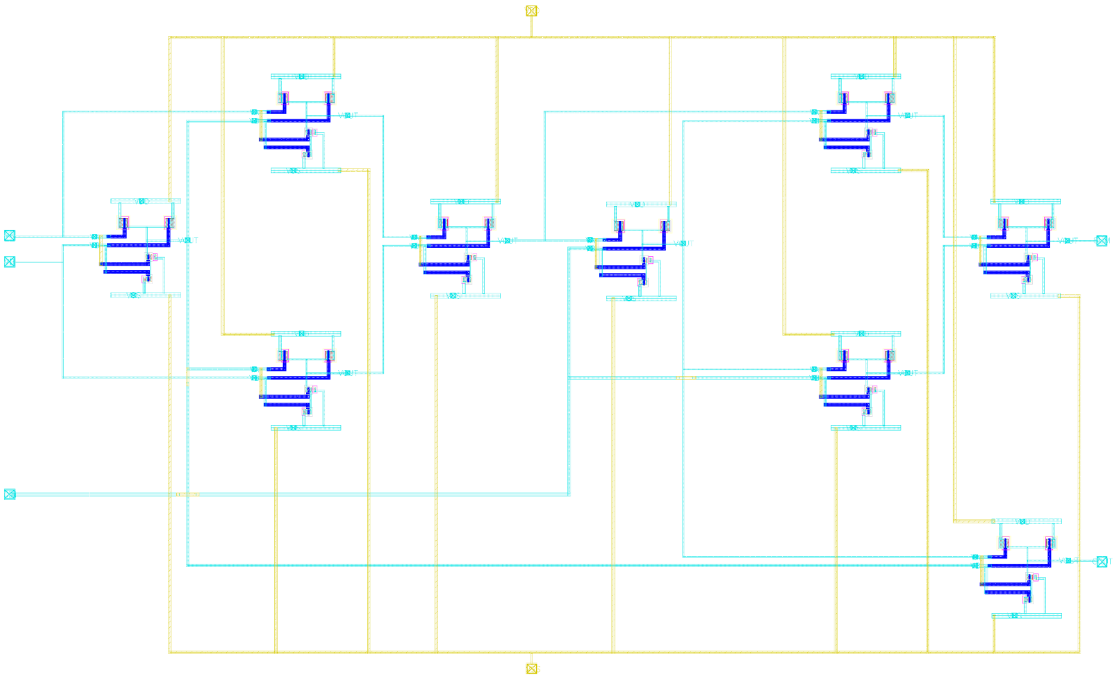


Figure 8: Full Adder Layout

The extracted layout of the full adder introduces several parasitic effects due to the larger number of interconnected NAND stages. The Calibre PEX results show that the highest parasitic capacitances occur at nodes that drive multiple logic inputs (shown in Figure 9). In particular, the outputs of the first and fifth NAND gates each drive three downstream gate inputs, resulting in larger metal–metal and metal–poly capacitances along these nets. These parasitic loads increase the effective capacitance seen by the driving transistors and contribute to the longer propagation delay observed along the Sum path, which naturally contains the highest number of cascaded stages.

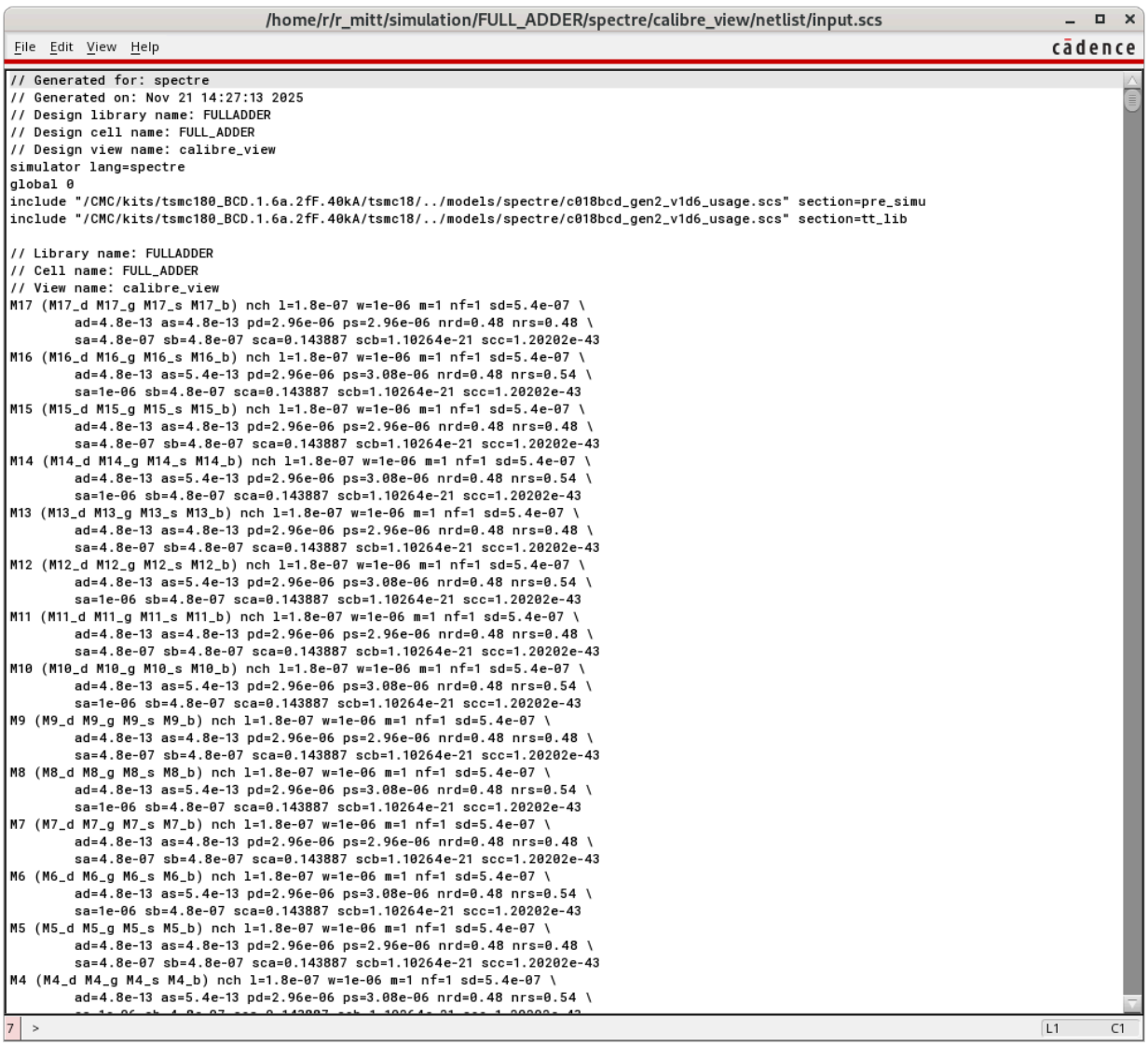


Figure 9: Full Adder PEX Extracted Netlist

Routing complexity varies across the layout, with longer interconnect segments found along the paths responsible for generating the Sum output. These nets require horizontal and vertical routing through multiple logic stages, leading to increased RC parasitics compared to shorter, local connections. Metal 1 and Metal 2 are used throughout the design to manage routing density and prevent congestion, while maintaining reasonably short and direct interconnects to limit additional RC delay. The Carry-generation logic, which involves fewer cascaded NAND gates, exhibits shorter routing length and correspondingly lower interconnect parasitics.

As in the NAND layout, diffusion regions inside each NAND gate are kept isolated rather than shared between devices, even in cases where shared diffusion would have been possible. This simplifies placement and routing and maintains consistently sized active regions across all nine NAND cells. Diffusion sharing does not occur *between* NAND gates due to differing connectivity requirements, so each gate retains its own distinct active regions and contacts. The regular placement of the cells and alignment of their boundaries help the overall layout remain compact despite the absence of shared diffusion structures.

The full-adder layout is area efficient given the nine gate structure. All NAND cells share common VDD and VSS rails, which run horizontally across the width of the design. Each gate connects to the VDD and VSS rails through single diffusion contacts and the required via stack for routing, providing a reliable power connection within the cell. Signal routing between gates is handled using Metal 1 and Metal 2, with via stacks inserted only where necessary for pin access or to transition between layers. This structured approach to routing and power distribution results in a compact and organized layout that supports efficient tiling and integration within larger arithmetic blocks.

The completed layout passed all required design rule checks (DRC), confirming proper spacing, enclosure, and device isolation. Layout-versus-schematic (LVS) verification also reported a perfect match between the physical layout and the intended full-adder schematic. Parasitic extraction (PEX) completed successfully, and the extracted view

was used for all post-layout simulations to ensure accuracy in the reported timing and power characteristics. (DRC and LVS confirmation shown in Appendix A.)

3 - Simulation Results

Both the NAND gate and the full adder design were simulated with an output load. The performance of the extracted circuits were compared to the performance achieved in [1] and to that of their schematics.

3.1 - NAND Gate

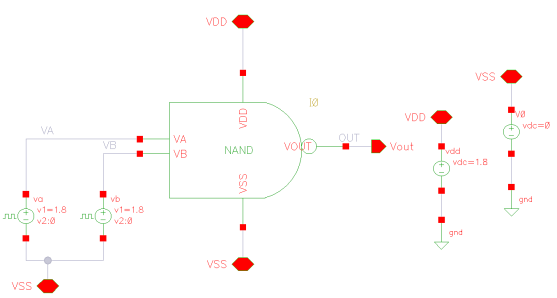


Figure 10: NAND Gate Simulation Testbench without Load

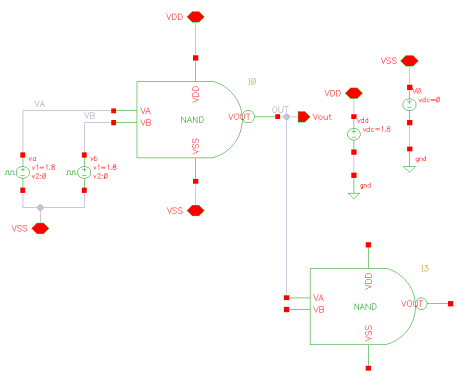


Figure 11: NAND Gate Simulation Testbench with Load

To obtain realistic propagation delay values for the NAND gate, an additional NAND gate was connected to its output during simulation as shown in Figure 11. This NAND gate serves as a capacitive load representative of the conditions present in the full-adder, where several gates must drive the inputs of other NAND stages. Since the input terminals of a NAND gate present a significant gate capacitance, loading the device under test in this way provides a more accurate estimate of the rise and fall delays compared to an unloaded configuration.

Table 2: NAND Gate Specifications

	Schematic		Layout	
	No load	With load	No load	With load
Delay to Out (ps)	-	30.79	40.28	40.28
$P_{dynamic,\,avg}(\mu W)$	-	1.49	1.95	1.95
$P_{static,\,avg}(\mu W)$	-	-	26.88	-
V_{OL} (V)	-	-	$20.98n$	-
V_{OH} (V)	-	-	1.8	-

3.1.1 - Propagation Delay

Figure 12 shows the 2-input NAND gate simulation waveform. The NAND gate performed its expected function, with V_A simulated with a 20ns period voltage pulse and V_B simulated with a 10ns period pulse. The delay from Edge 4 of the V_B pulse and Edge 2 of output was measured to give an approximate propagation delay of the NAND gate.

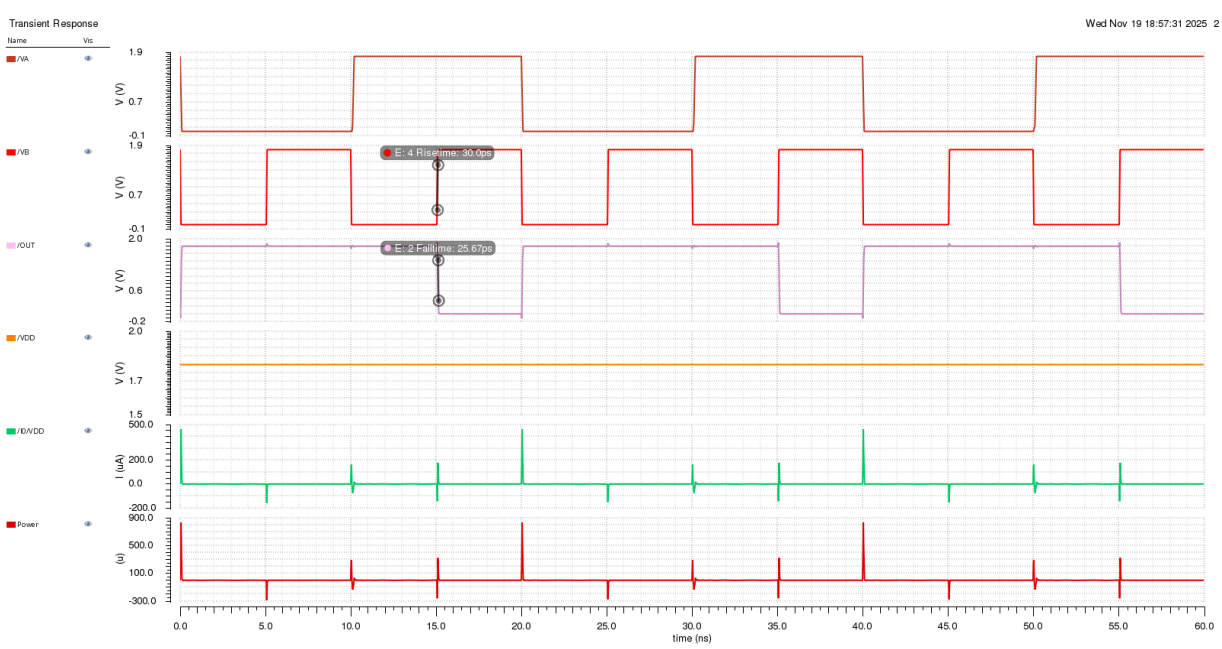


Figure 12: NAND Gate Simulation Waveform

Delay from V_B to output came out to be ~ 40.28ps in the extracted NAND circuit whereas in non-extracted NAND testbench, the delay was ~30.8ps.

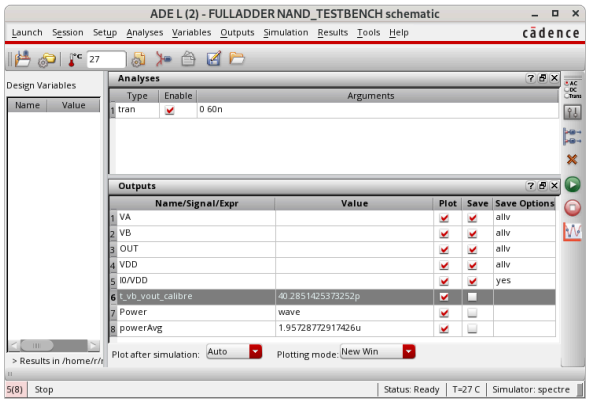


Figure 13: Delay and Average Power of NAND Layout

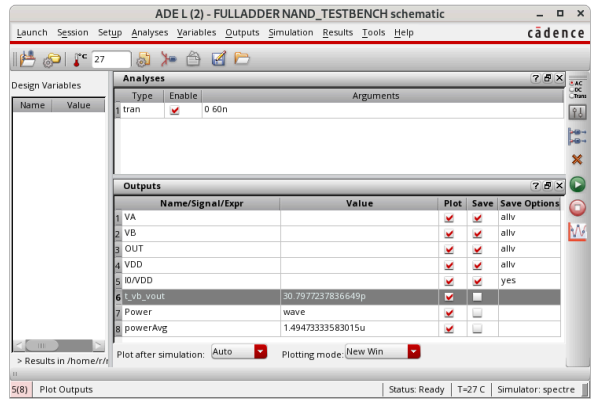


Figure 14: Delay and Average Power of NAND Schematic

3.1.2 - Power Dissipation

The gate's dynamic power dissipation peaks when both PMOS are turned on (V_A = V_B = 0V) and the output is initially at 0V (see Figure 12). At that instant, the two parallel PMOS transistors supply the maximum charging current to the output capacitance, causing the highest instantaneous dynamic power dissipation.

Since the NAND gate uses a CMOS configuration, it consumes very little static power (~37.5pW) when the inputs are ideal (near 0 or V_{DD}) as shown in Figure 15. In this case, either the pull-up or the pull-down network is fully off and leakage currents consume a small amount of power.

When an input (V_A in this DC simulation) is not ideal, the transistors in the pull-up and pull-down networks are partially on, resulting in short circuit power dissipation. This

short circuit power peaks at 133μW when the input is 864mV (the point where the pull-up and pull-down networks are most conductive, as seen in Figure 15.

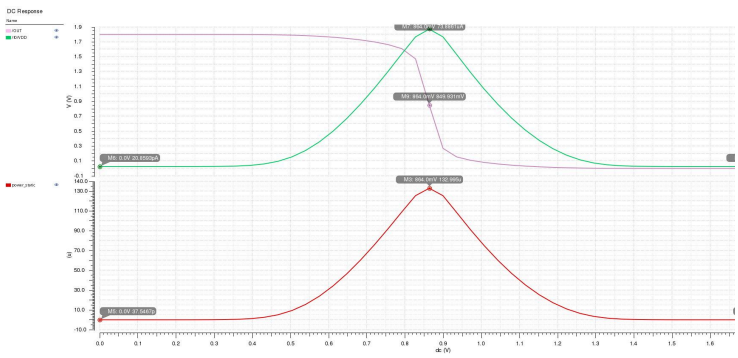


Figure 15: NAND Gate Layout Static Power Dissipation

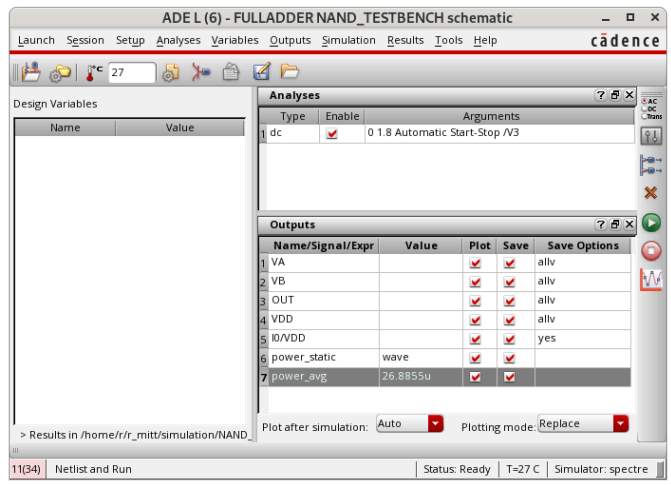


Figure 16: NAND Gate Layout DC Simulation Outputs

3.1.3 - Voltage Levels

The NAND gate produced $V_{OL} = 20.98nV$ and $V_{OH} = 1.8V$ with input VB DC sweeping from 0 to 1.8V and VA remaining constant at 1.8V. These strong voltage levels are expected in CMOS configurations.

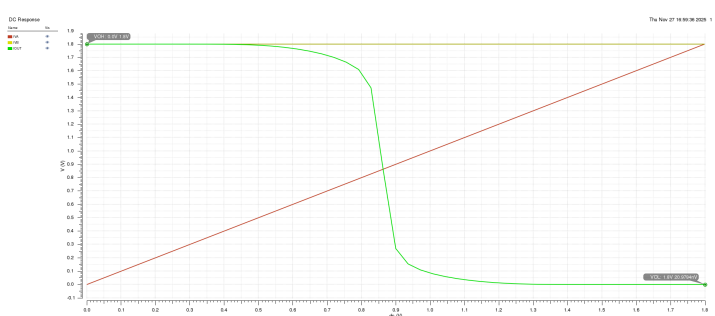


Figure 17: NAND Gate Voltage Levels

3.2 - Full Adder

The simulated full-adder design performed the expected 1-bit addition operation. The simulation of the extracted layout experienced longer delays and higher power dissipation due to parasitic capacitances and resistances.

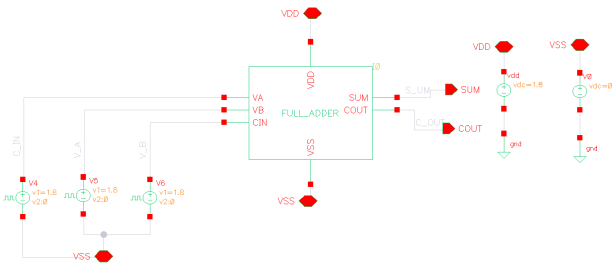


Figure 18: Fulladder Simulation Testbench without Load

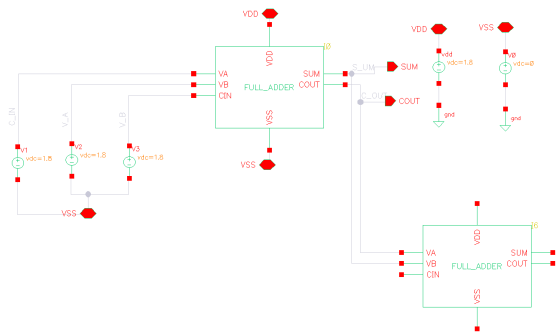


Figure 19: Fulladder Simulation Testbench with Load

To obtain realistic propagation delay and power measurements for the full adder, an additional full-adder cell was connected to the primary output during simulation, as shown in Figure 19. This configuration provides a representative capacitive load, since in practical digital systems the output of a 1-bit adder commonly drives the input of another adder stage in a ripple-carry architecture as mentioned in [1]. The input pins of the downstream full adder introduce substantial gate capacitance, and loading the circuit under test with an identical cell ensures that the simulated delays reflect the actual fan-out conditions encountered in a multi-bit implementation.

By applying this loading strategy, the transition times and dynamic power obtained from post layout simulation incorporate not only the parasitic capacitances extracted from the layout, but also the capacitive effects of the next arithmetic stage. This yields more accurate and meaningful timing and power results compared to an unloaded or artificially loaded configuration, and aligns the simulation environment with the intended usage of the full-adder design.

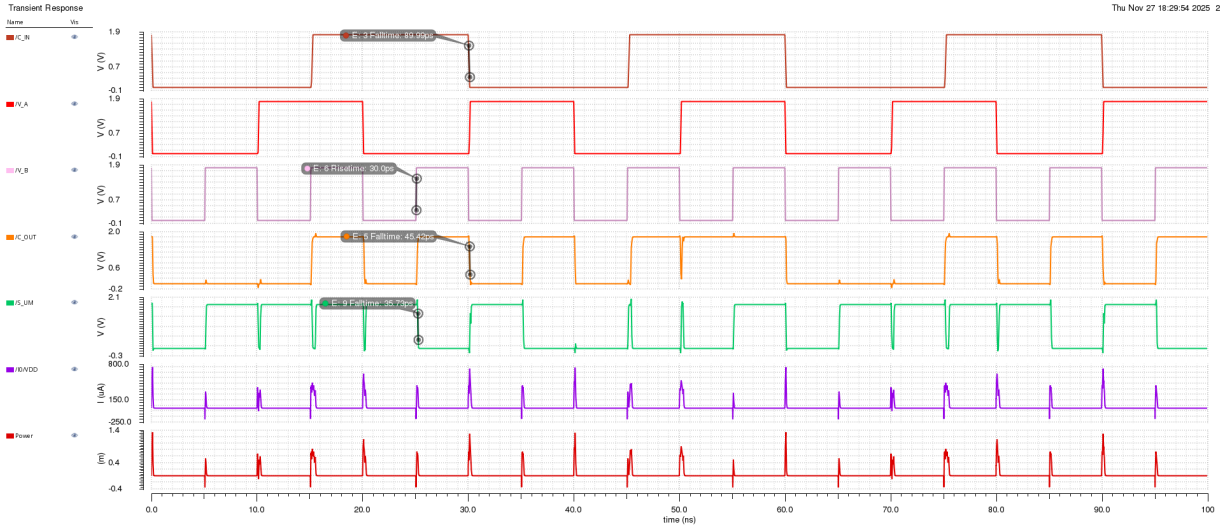


Figure 20: Full Adder Schematic Simulation

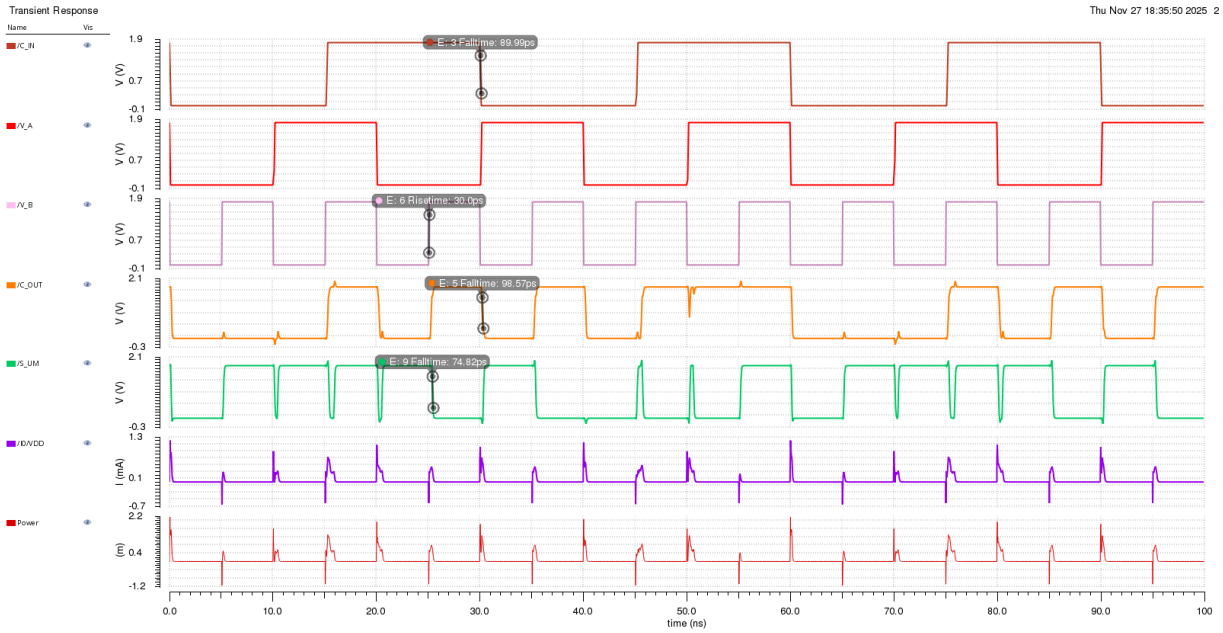


Figure 21: Full Adder Extracted Simulation

Table 3 summarises the full adder specifications, which will be elaborated upon in the following sections. The delay in these specifications is the average propagation delay from an input to the Sum output.

The delays achieved in [1] were given in ns, which is assumed to be a mistake given the results in other literature [2]. Furthermore, the delay for the full adder schematic was less than half that of a similar NAND-based adder in 90nm technology evaluated in [2]. The validity of the data in [1] is therefore uncertain.

Table 3: Full Adder Specifications

	Schematic		Layout	
	No load	With load	No load	With load
Delay to Sum (ps)	142.3	161.6	298.9	353.5
Delay to C_{OUT}(ps)	42.37	53.43	144.146	179.46
$P_{dynamic,\,avg}(\mu W)$	27.94	31.27	53.85	62.62
$P_{static,\,avg}(\mu W)$	-	-	45.94	-
Silicon area (μm^2)	-	-	26007.59	-
V_{OL} (V)	-	-	28.98n	-
V_{OH} (V)	-	-	1.8	-

3.2.1 Propagation Delay

As expected, the path to the Sum output experienced the longest propagation delay and was most impacted by the layout parasitics.The delays measured from the full adder design shall be compared to those of the reference design [1] in Section 3.3.

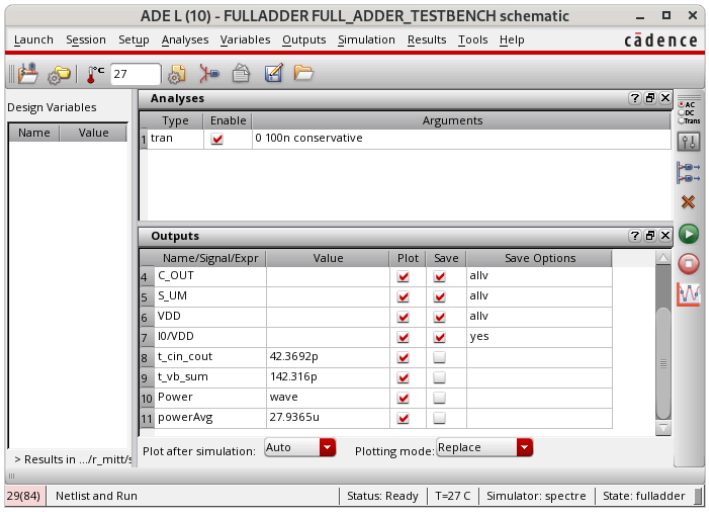


Figure 22: Average Power and Delay of Full Adder Schematic (Without Output Load)

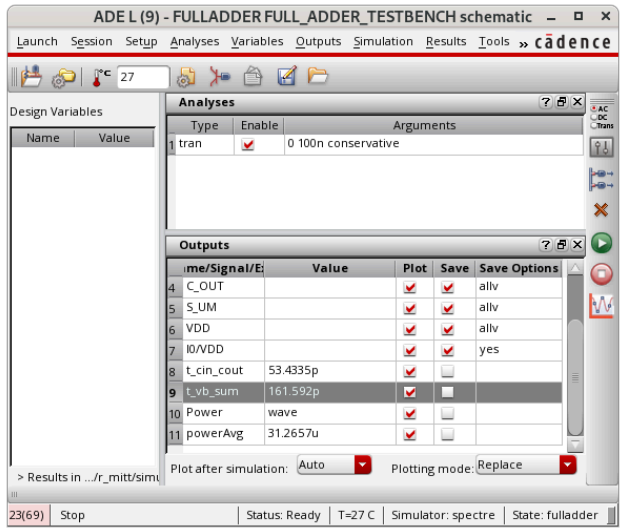


Figure 23: Average Power and Delay of Full Adder Schematic (With Output Load)

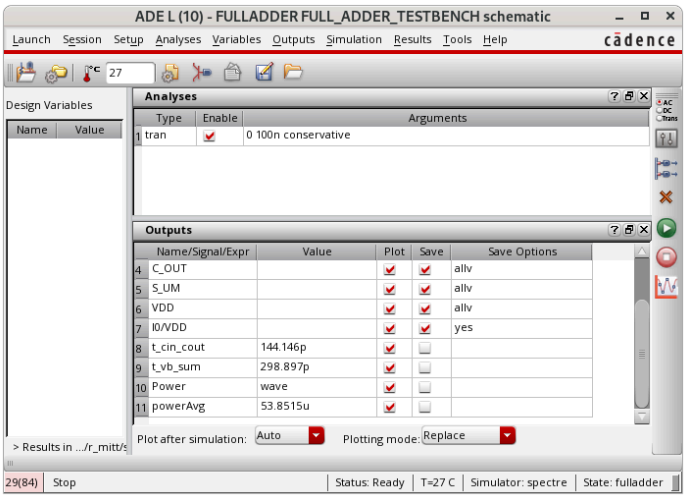


Figure 24: Average Power and Delay of Full Adder Layout (Without Output Load)

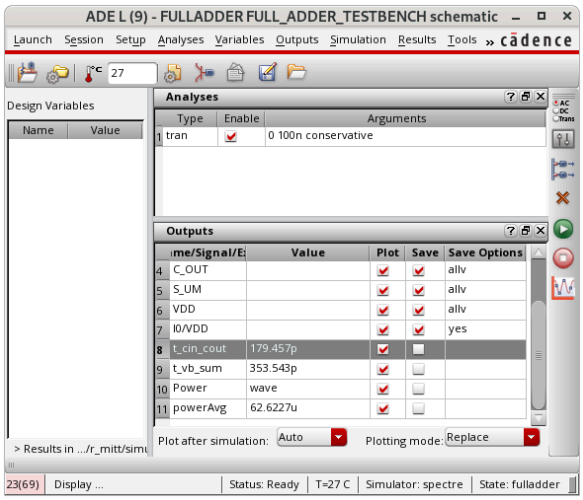


Figure 25: Average Power and Delay of Full Adder Layout (With Output Load)

3.2.2 - Power Dissipation

Dynamic power dissipation in the full adder was evaluated using post-layout transient simulation of the extracted view. The simulator reports an average dynamic power of 62.6 μW under the applied input transitions (Figure 25). This represents the energy required to charge and discharge the capacitive loads associated with internal nodes, the outputs of each NAND stage, and the additional parasitic capacitances introduced by the layout. As discussed in Section 2.2, several nodes, particularly the outputs of the first and fifth NAND gates drive three gate inputs, resulting in higher effective load capacitance. Each switching event on these nets therefore contributes more dynamic energy, increasing overall power consumption.

The Sum generation path has the most cascaded gates; multiple internal nodes toggle when a primary input is changed. This higher switching activity factor further increases the dynamic power contribution from this portion of the circuit. Since the simulation uses the extracted netlist, both interconnect RC parasitics and diffusion capacitances are included in the power calculation, yielding a realistic estimate of the full adder's dynamic power under typical operating conditions.

Similar to the isolated NAND gate simulation, the adder consumes very little static power when the inputs are ideal.
When an input (C_{in} in this DC simulation) is not ideal, then all gates tied to that input are partially on, resulting in short circuit power dissipation. This short circuit power peaks at 498.14μW when the input is 864mV as shown in Figure 27.. In the DC simulation, V_A = V_B = V_{DD} and C_{in} was swept from 0 to 1.8V. This DC sweep resulted in a 45.94μW average static power dissipation, as shown in Figure 26.

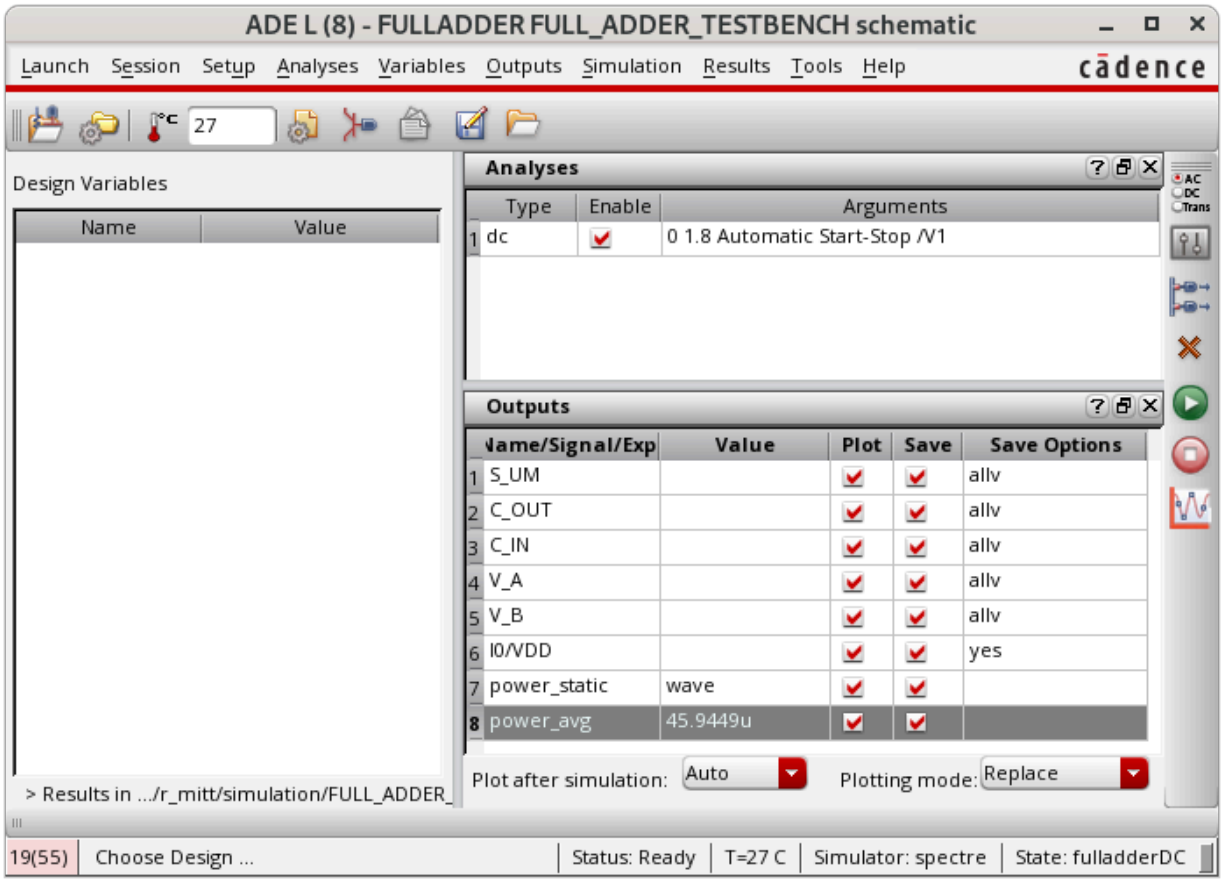


Figure 26: Full Adder DC Simulation Results

3.2.3 Voltage Levels

In the DC simulation, $V_A = V_B = V_{DD}$ and C_{in} was swept from 0V to V_{DD}. The full adder design had strong voltage levels of $V_{OL} = 21.98nV$ and $V_{OH} = 1.8V$ (see Figure 27), as is expected for CMOS technology.

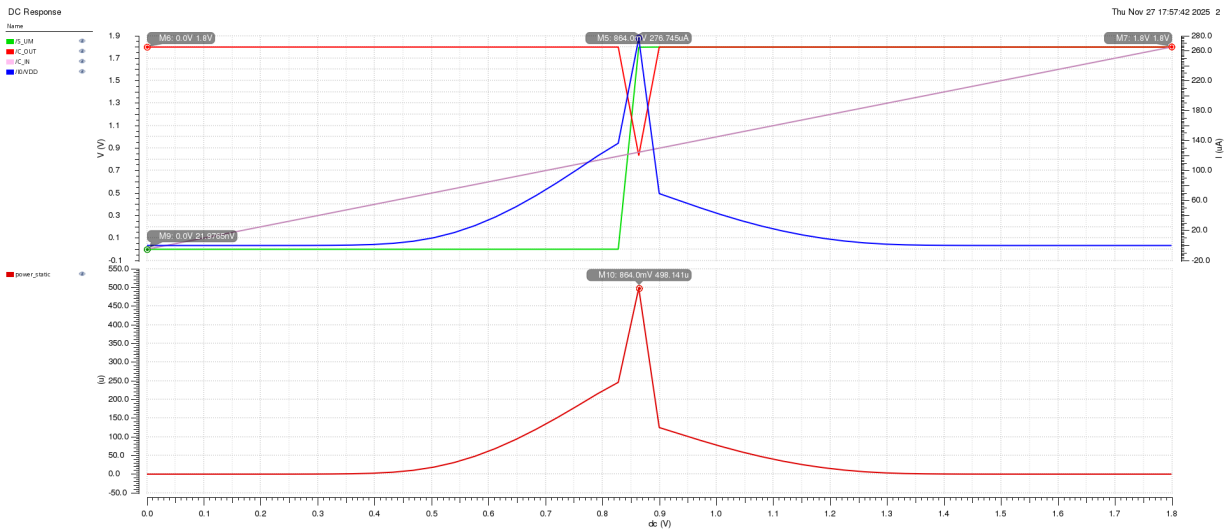


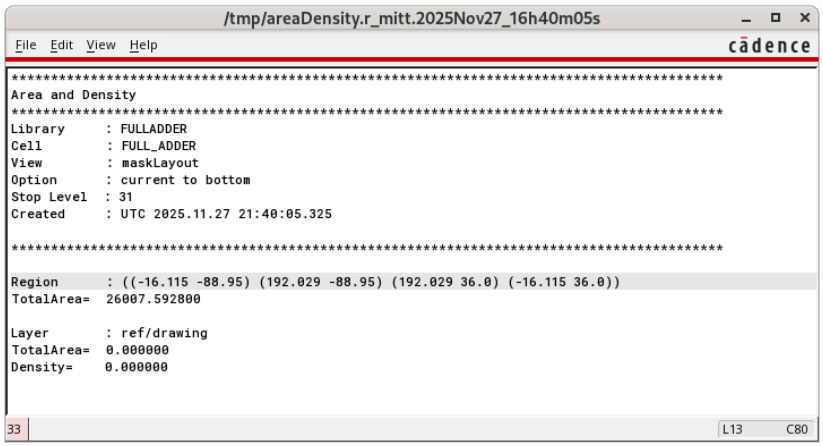
Figure 27: Full Adder Voltage Levels

3.2.4 - Silicon Area

The full adder layout occupied a silicon area of 26007.59μm^2. The adder consists of 9 4T NAND gates; a total of 36 transistors. The large silicon area can be attributed to the use of individually placed transistor cells, excessive spacing between transistors and between NAND gates, and long metal interconnects.

A more experienced layout designer would use minimal spacing between elements, shared diffusion, and efficient signal routing to optimize their design for area efficiency.

Area inefficiency is a significant oversight in this design and explains the long propagation delay times and large power consumption relative to optimised adder designs seen in [1][2].



```
/tmp/areaDensity.r_mitt.2025Nov27_16h40m05s
File  Edit  View  Help                                                    cadence
*****************************************************************************************
Area and Density
*****************************************************************************************
Library      : FULLADDER
Cell         : FULL_ADDER
View         : maskLayout
Option       : current to bottom
Stop Level   : 31
Created      : UTC 2025.11.27 21:40:05.325

*****************************************************************************************

Region       : ((-16.115 -88.95) (192.029 -88.95) (192.029 36.0) (-16.115 36.0))
TotalArea=   26007.592800

Layer        : ref/drawing
TotalArea=   0.000000
Density=     0.000000

33                                                                    L13     C80
```

Figure 28: Full Adder Silicon Area

3.3 - Comparison to the Performance in Reference Design [1]

The average power consumption of the NAND gate and Full Adder design, calculated during the transient analysis of the extracted layout, was very similar to the power consumption achieved in [1].

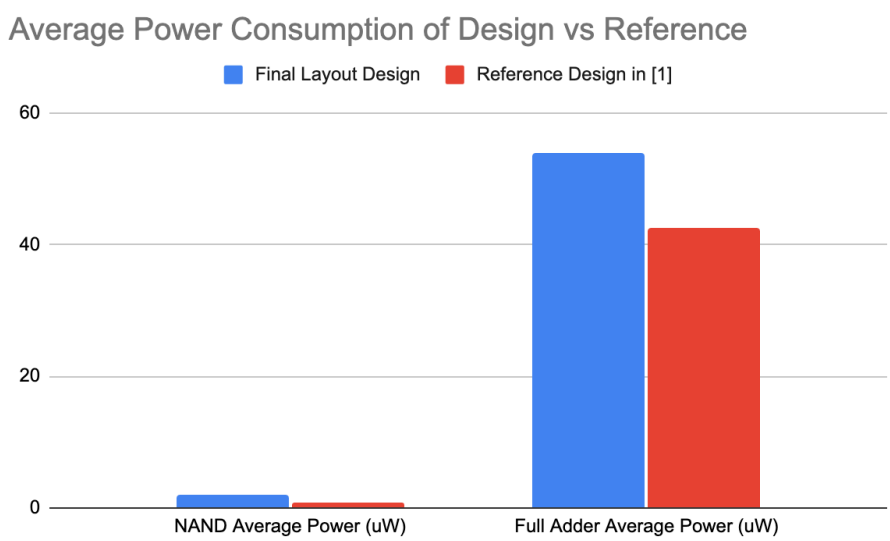


Figure 29: Comparing Power Consumption to Reference Design

The delays calculated for the NAND gate were close, but not identical, to those given in [1]. The schematic designs should have identical delays, as the MOSFETs were sized the same; the difference in delays is likely due to a difference in how delay was calculated in the reference design. The difference in delays between the layout design in this project and that achieved in [1] is expected, since the two layouts introduce different parasitics.

The delays calculated for the full adder design were much greater than those given in [1]. The delay value given for the full adder schematic design in [1] is likely incorrect, as it is much less than that of their own NAND gate delay and less than half of the delay achieved by the 90nm NAND-based full adder in [2]. The layout design delay presented in [1], however, is more reasonable when compared to the results of [2]. The delays obtained from the full adder layout designed in this project were significantly greater than expected and could be improved by optimising the NAND gate layout, shortening interconnects, and redesigning the power routing paths.

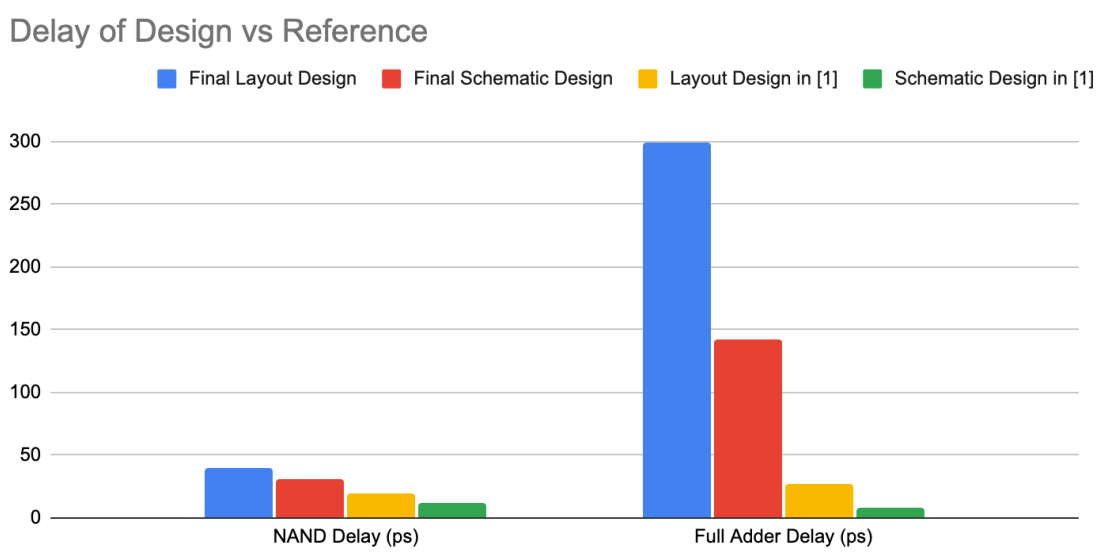


Figure 30: Comparing Delay to Reference Design

4 - Conclusion

The average dynamic power consumption of the NAND gate and full-adder obtained from transient simulations closely matched the results in [1], as was expected given the identical transistor sizing used in both designs.

The propagation delays measured from the NAND gate and full adder design were significantly larger than those achieved in [1]. The discrepancy in the NAND gate delay can be attributed to differences in delay measurement methodologies. The delays reported for the full adder schematic in [1] are likely incorrect, as discussed in Section 3.3, but those of the layout provide a realistic expected performance. The delays measured for the full adder layout in this project were substantially larger than those in [1], likely due to layout design oversights such as unnecessary spacing and long paths routing the gates to the power supply and ground.

The layout design could be revised to improve the full adder propagation delays; in particular by replacing the hand-drawn NAND gates with standard cells, shortening connections between cascaded gates, adding contacts to the gates' sources to reduce resistance, and routing power and input signals more efficiently.

References

[1] M. C. Trupthi, D. Savitha, "Design Of Ripple Carry Adder Using Cadence Virtuoso 180nm Technology," in *2025 3rd International Conference on Smart Systems for applications in Electrical Sciences (ICSSES)*, Tumakuru, India, 2025.

[2] H. N. Y. Pwint and T. T. Hla, "Performance Analysis of Full Adder Using Different CMOS Technology," 2024 IEEE Conference on Computer Applications (ICCA), Yangon, Myanmar, 2024, pp. 1-5, doi: 10.1109/ICCA62361.2024.10532847.

Appendix A: NAND and Full Adder Layout DRC and LVS Results

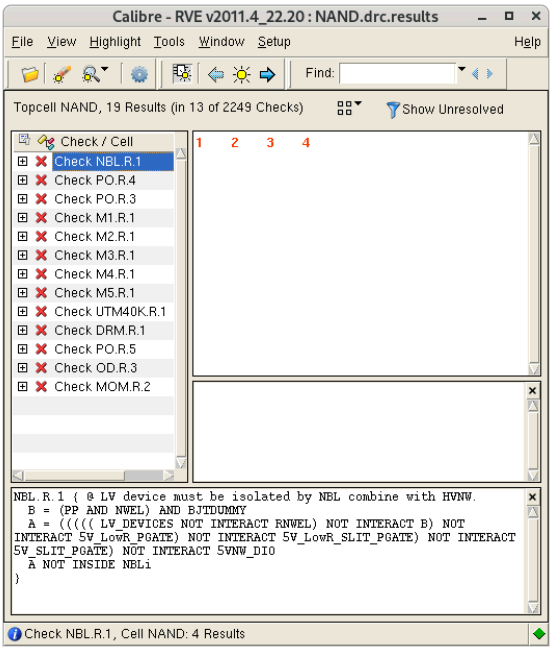


Figure 31: NAND Layout DRC Results

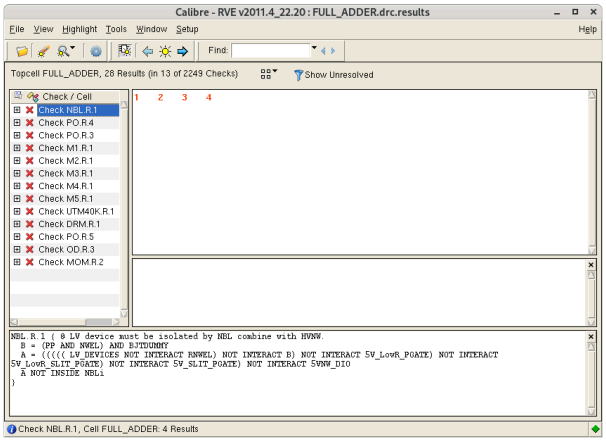


Figure 32: Full Adder Layout DRC Results

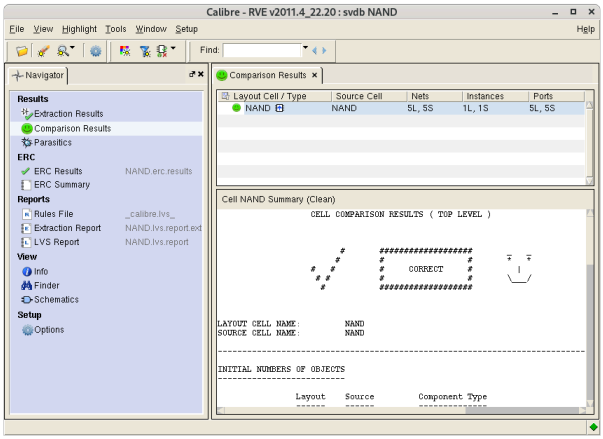


Figure 33: NAND Layout LVS Results

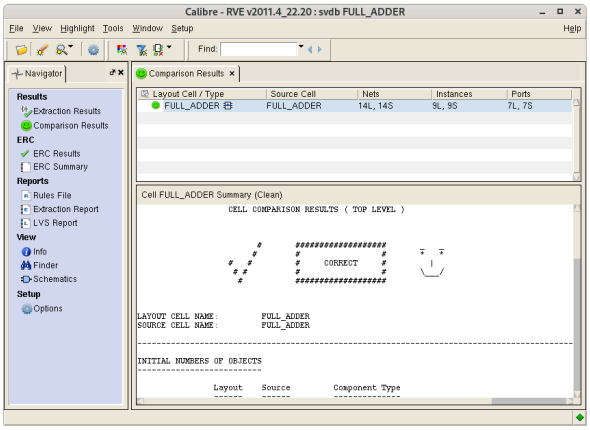


Figure 34: Full Adder Layout LVS Results